鄧惠文 Huei-Wen Teng

Department of Information Management and Finance
National Yang Ming Chiao Tung University
No. 1001, Daxue Rd. East Dist.
Hsinchu 300093, Taiwan, ROC
Phone: +886-3 571 2121 ext 57092
E-mail: hwteng@nycu.edu.tw

Web: https://venteng.github.io
Google Scholar: HDPfo-IAAAAJ
254 citations, h-index 9 (Jul 2026)
GitHub: venteng

PROFILE

Professor of quantitative finance and FinTech; founder and director of the FIMA Lab (est. 2010). Author of 27 refereed journal articles — including in *International Review of Financial Analysis* (IF 10.2), *Finance Research Letters* (IF 7.1), and *Financial Innovation* (IF 6.4) — with 8 papers currently under review. NSTC Principal Investigator on 17 projects since 2010 (NTD 15.5 million), and Co-Principal Investigator on projects totaling NTD 15.0 million. Three best-paper awards, two patents, and over 40 graduate theses supervised. 2026 invited talks include the European Central Bank (Frankfurt), the National University of Singapore, and the AIDA Conference (Bucharest).

ACADEMIC EXPERIENCES

National Yang Ming Chiao Tung University (formerly National Chiao Tung University; merged in Feb 2021)

Professor, Department of Information Management and Finance	Aug 2024 – Present
Associate Professor, Department of Information Management and Finance	Feb 2017 – Jul 2024
Jointly Appointed Position, Institute of Statistics	Aug 2017 – Present
Jointly Appointed Position, Department of Applied Mathematics	Oct 2024 – Present

FIMA Lab (Finance · Intelligence · Math · Analytics)

Founder and Director	2010 – Present

IDA Institute Digital Assets, The Bucharest University of Economic Studies, Romania

Experienced Researcher	Apr 2024 – Present

Humboldt-Universität zu Berlin, Germany

Visiting Scholar	Summers 2011, 2012, 2023

National Central University

Associate Professor, Graduate Institute of Statistics	Aug 2016 – Jan 2017
Jointly Appointed Position, Department of Finance	Aug 2016 – Jan 2017
Assistant Professor, Graduate Institute of Statistics	Aug 2010 – Jul 2016

EDUCATION

Ph.D. in Statistics, The Pennsylvania State University — Jul 2010
Dissertation: *Bayesian Nonparametric Approaches for Financial Option Pricing*

MBA in Finance, National Taiwan University — Jul 2004

BS in Mathematics, National Taiwan University — Jul 2001

Exchange Student, Johannes Kepler University, Austria — Aug 1999 – Jan 2000

RESEARCH INTERESTS

Explainable AI and FinTech: machine learning for credit scoring and default prediction

Trading and portfolio management: empirical asset pricing with machine learning

Financial market risk management: tail risk and digital asset regulation

Financial derivatives pricing: Monte Carlo simulation and cryptocurrency options

AWARDS

1. "Financial Analytics of Deribit BTC Options" (with Wolfgang Karl Härdle), The 31st Annual Conference on Pacific Basin Finance, Economics, Accounting and Management (PBFEAM), Best Paper Award (one of ten, USD 1,000) 第三十一屆亞太財務經濟會計與管理研討會, 最佳論文 Jun 2, 2023
2. "透過群表現理論的一個新穎的球狀蒙地卡羅估計量來計算高維度積分" (鄧惠文、康明軒、李潤澤) 第18屆台灣作業研究學會年會暨學術研討會(ORSTW 2022), 最佳論文 Dec 9, 2022
3. "比特幣倒數型選擇權在隨機波動度下的動態避險" (張詠淇、鄧惠文、沃夫岡・赫德勒) 2022 New Futures 期貨學術與實務交流研討會, 臺灣期貨交易所與工商時報共同舉辦, Taipei, 論文金質獎Nov 28, 2022
4. 指導研究生 邱翰奕 (Han-Yi Chiou) 獲 2026 第19屆崇越論文大賞 佳作論文獎；證書頒予 鄧惠文 教授，嘉惠後學、指導卓著 (台灣管理學會) Sep 22, 2026
5. Student awards under supervision: 中國統計學社論文獎優等 (陳醇潔 Chun-Chieh Chen, 2012); NYCU 第一屆智慧創新跨域潛力人才競賽金融科技組, 優等 × 2 — 陳之怡、唐君亞、蔡昕妤「台灣高市值電子與金融行業的金融風險計量指標」and 王鳴謙、吳珮慈、李語涵、鄭有朋、許晉「探索比特幣回報分析的多面向驅動因素」(Dec 15, 2023) — and 優選 (2025); NSTC undergraduate research grant MOST109-2813-C-009-069-H (林睿瑜 Rui-Yu Lin, 2020)

PUBLICATIONS

Refereed Academic Journal Articles (27)

1. **Huei-Wen Teng** and Han-Yi Chiou. Forecasting intraday Bitcoin volatility: Do financial news embeddings add value beyond market microstructure? *Finance Research Letters*, October 2026. URL https://venteng.github.io/papers/teng2026forecasting-AAM.pdf. Accepted, in production
2. Hung-Yin Chen, Henghsiu Tsai, PingHan Chen, and **Huei-Wen Teng**. AI Act compliance in credit scoring: reconciling interpretability and accuracy via quadratic terms and powershap. *Applied Economics Letters*, pages 1–7, August 2026. doi:10.1080/13504851.2026.2710263

3. **Huei-Wen Teng**, Yu-Chuan Shih, and Ming-Hsuan Kang. Qualitative insights meet quantitative forecasts with GPT-4o mini for Taiwan earnings prediction. *Journal of the Chinese Statistical Association*, 64:417–450, 2026b. URL https://venteng.github.io/papers/teng2026qualitative-AAM.pdf. (EconLit)

4. Ioana Florina Coita, Marcos R. Machado, Lucia Gomez Teijeiro, Karsten Wenzlaff, Andreas Gregoriades, Christos Themistocleous, Frédérik Sinan Bernard, Ramona Rupeika-Apoga, **Huei-Wen Teng**, Anastas Dzurovski, Gokce Nur Yilmaz, Liana Stanca, Wouter van Heeswijk, Marius Vlad Pop, Karolina Bolesta, Jana Peliova, Olivija Filipovska, and Joerg Osterrieder. Taxonomy of fraud types in alternative finance using hybrid systematic review. *International Journal of Information Management Data Insights*, 6(2):100435, 2026. ISSN 2667-0968. doi:10.1016/j.jjimei.2026.100435. URL https://www.sciencedirect.com/science/article/pii/S2667096826000480

5. **Huei-Wen Teng**, Wolfgang K Härdle, Jörg Osterrieder, Daniel Traian Pele, and et al. Digital assets: risks, regulations, mitigation. *Financial Innovation*, 12(1):65, February 2026a. doi:10.1186/s40854-025-00848-y

6. **Huei-Wen Teng**, Hsin-Pei Huang, and Yu-Chuan Shih. Tail risk in Bitcoin under the Basel framework. *Finance Research Letters*, 86:108528, December 2025. doi:10.1016/j.frl.2025.108528

7. Yu-Jui Chen, Chen-Yen Hsieh, **Huei-Wen Teng**, Ming-Che Hu, and Alex YiHou Huang. Risk parity strategies with risk factors. *Journal of Risk*, 27(5):1–27, 2025. doi:10.21314/JOR.2025.006

8. Le Quoc Tuan, Chih-Yung Lin, and **Huei-Wen Teng**. Machine learning methods for predicting failures of US commercial bank. *Applied Economics Letters*, 31(15):1353–1359, 2024. doi:10.1080/13504851.2023.2186353

9. **Huei-Wen Teng**, Ming-Hsuan Kang, I-Han Lee, and Le-Chi Bai. Bridging accuracy and interpretability: A rescaled cluster-then-predict approach for enhanced credit scoring. *International Review of Financial Analysis*, 91:103005, January 2024. doi:10.1016/j.irfa.2023.103005. URL https://venteng.github.io/papers/teng2024bridging-AAM.pdf

10. **Huei-Wen Teng**. Importance sampling for calculating the value-at-risk and expected shortfall of the quadratic portfolio with t-distributed risk factors. *Computational Economics*, 62(3):1125–1154, October 2023. doi:10.1007/s10614-022-10294-y. URL https://venteng.github.io/papers/teng2023importance-AAM.pdf

11. Yung-Chi Chang, **Huei-Wen Teng**, and Wolfgang Härdle. Stochastic volatility dynamic hedging for Deribit BTC options. *Journal of Futures and Options*, 16(2):1–48, August 2023. URL https://www.taifex.com.tw/cht/10/journalView. (TSSCI; no DOI registered. Working-paper version: doi 10.2139/ssrn.4271856)

12. **Huei-Wen Teng** and Yu-Hsien Li. Can deep neural networks outperform Fama-Macbeth regression and other supervised learning approaches in stock returns prediction with asset-pricing factors? *Digital Finance*, 5(1):149–182, March 2023. doi:10.1007/s42521-023-00076-y. URL https://venteng.github.io/papers/teng2023can-AAM.pdf

13. **Huei-Wen Teng** and Cheng-Der Fuh. Simulating false alarm probability in K-distributed sea clutter. *Communications in Statistics-Simulation and Computation*, 51(9):5081–5098, September 2022. doi:10.1080/03610918.2020.1757707

14. **Huei-Wen Teng** and Ming-Hsuan Kang. On accelerating Monte Carlo integration using orthogonal projections. *Methodology and Computing in Applied Probability*, 24(2):1143–1168, June 2022. doi:10.1007/s11009-021-09893-3. URL https://venteng.github.io/papers/teng2022accelerating-AAM.pdf

15. **Huei-Wen Teng** and Xiang-An Zhao. On pricing quanto options with spherical Monte Carlo simulation. *Journal of Futures and Options*, 14(2):25–82, August 2021. doi:10.6935/JFO.202108_14(2).0002. (TSSCI)

16. **Huei-Wen Teng** and Michael Lee. Estimation procedures of using five alternative machine learning methods for predicting credit card default. *Review of Pacific Basin Financial Markets and Policies*, 22 (03):1950021, September 2019. doi:10.1142/S0219091519500218

17. Yuh-Dauh Lyuu, **Huei-Wen Teng**, Yao-Te Tseng, and Sheng-Xiang Wang. A systematic and efficient simulation scheme for the Greeks of financial derivatives. *Quantitative Finance*, 19(7):1199–1219, January 2019. doi:10.1080/14697688.2018.1562196

18. Cheng-Der Fuh, **Huei-Wen Teng**, and Ren-Her Wang. Efficient simulation of value-at-risk under a jump diffusion model: A new method for moderate deviation events. *Computational Economics*, 51(4): 973–990, April 2018. doi:10.1007/s10614-017-9654-z

19. **Huei-Wen Teng**, Cheng-Der Fuh, and Chun-Chieh Chen. On an automatic and optimal importance sampling approach with applications in finance. *Quantitative Finance*, 16(8):1259–1271, March 2016. doi:10.1080/14697688.2015.1136077

20. Sheng-Feng Luo, **Huei-Wen Teng**, and Yu-Hsuan Lee. Forecasting mortality using imputed data: The case of Taiwan. *Asia-Pacific Journal of Risk and Insurance*, 10(1):1–20, 2016. doi:10.1515/apjri-2015-0011

21. **Huei-Wen Teng**, Ming-Hsuan Kang, and Cheng-Der Fuh. On spherical Monte Carlo simulations for multivariate normal probabilities. *Advances in Applied Probability*, 47(3):817–836, September 2015b. doi:10.1239/aap/1444308883

22. Wolfgang Karl Härdle, Brenda López-Cabrera, and **Huei-Wen Teng**. State price densities implied from weather derivatives. *Insurance: Mathematics and Economics*, 64:106–125, September 2015. doi:10.1016/j.insmatheco.2015.05.001

23. **Huei-Wen Teng**, Wen-Liang Hung, and Yen-Ju Chao. Bayesian Markov chain Monte Carlo imputation for the transiting exoplanets with an application in clustering analysis. *Journal of Applied Statistics*, 42(5):1120–1132, 2015a. doi:10.1080/02664763.2014.995609

24. Chun-Chieh Chen, Cheng-Der Fuh, and **Huei-Wen Teng**. Efficient option pricing with importance sampling. *Journal of the Chinese Statistical Association*, 51(3):253–273, September 2013. URL `https://www.airitilibrary.com/Article/Detail/05296528-201309-201309020004-201309020004-253-273`. (EconLit; no DOI registered)

25. Yuh-Dauh Lyuu and **Huei-Wen Teng**. Unbiased and efficient Greeks of financial options. *Finance and Stochastics*, 15(1):141–181, January 2011. doi:10.1007/s00780-010-0137-5

26. Cheng-Der Fuh, **Huei-Wen Teng**, and Ren-Her Wang. On-line VWAP trading strategies. *Sequential Analysis*, 29(3):292–310, July 2010. doi:10.1080/07474946.2010.487431

27. Tse-Chuan Yang, **Huei-Wen Teng**, and Murali Haran. The impacts of social capital on infant mortality in the U.S.: A spatial investigation. *Applied Spatial Analysis and Policy*, 2(3):211–227, June 2009. doi:10.1007/s12061-009-9025-9

Comments in Refereed Academic Journal Articles

1. Cheng-Der Fuh and **Huei-Wen Teng**. Discussion of "multiscale change point inference" by Frick, Munk and Sieling. *Journal of the Royal Statistical Society: Series B*, 76(3):554–555, 2014

Articles in Conference Proceedings

1. **Huei Wen Teng**, Yu-Hsien Li, and Shang-Wen Chang. Machine learning in empirical asset pricing models. In *2020 International Conference on Pervasive Artificial Intelligence (ICPAI)*, pages 123–129. IEEE, December 2020. doi:10.1109/ICPAI51961.2020.00030

2. **Huei-Wen Teng**. A spherical Monte Carlo approach for calculating value-at-risk and expected shortfall in financial risk management. In *2017 Winter Simulation Conference (WSC)*, pages 469–480. IEEE, December 2017. doi:10.1109/WSC.2017.8247808

Book Chapters (4)

1. **Huei-Wen Teng**. Comparisons between the Markowitz model and the Black-Litterman model. In *Handbook of Investment Analysis, Portfolio Management, and Financial Derivatives*, volume 3, chapter 85, pages 2727–2749. World Scientific, Singapore, 2024. doi:10.1142/9789811269943_0085

2. **Huei-Wen Teng** and Michael Lee. Alternative machine learning methods for credit card default forecasting. In John Lee, Jow-Ran Chang, Lie-Jane Kao, and Cheng Few Lee, editors, *Essentials of Excel VBA, Python, and R Volume II: Financial Derivatives, Risk Management and Machine Learning*, volume 2, chapter 14, pages 285–298. Springer, Switzerland, 2 edition, 2023. doi:10.1007/978-3-031-14283-3_14

3. Wei-Hung Lin, **Huei-Wen Teng**, and Chi-Chun Yang. A heteroskedastic Black-Litterman portfolio optimization model with views derived from a predictive regression. In Cheng Few Lee and John Lee, editors, *Handbook of Financial Econometrics, Mathematics, Statistics, and Machine Learning*, volume 1, chapter 14, pages 563–581. World Scientific, Hackensack, NJ, 2020. doi:10.1142/9789811202391_0014

4. **Huei-Wen Teng** and Michael Lee. Estimation procedures of using five alternative machine learning methods for predicting credit card default. In Cheng Few Lee and John Lee, editors, *Handbook of Financial Econometrics, Mathematics, Statistics, and Machine Learning*, volume 4, chapter 101, pages 3545–3572. World Scientific, Hackensack, NJ, 2020. doi:10.1142/9789811202391_0101

PATENTS

1. 鄧惠文,國立陽明交通大學。 評估投資組合風險指標方法及投資組合評估裝置 (*Method for evaluating risk measures for portfolio and portfolio evaluating device*)，中華民國經濟部智慧財產局，發明第I765347號，申請案號109131338，2020/09/11-2040/09/10。

2. 鄧惠文，康明軒，傅承德，國立中央大學。金融商品定價系統以及金融商品定價方法, 中華民國經濟部智慧財產局，發明第I534737號，2016/5/21-2035/3/26。

3. Huei-Wen Teng, National Yang Ming Chiao Tung University. *Method for evaluating risk measures for portfolio and portfolio evaluating device*. USPTO application 17/447,060, filed Sep 2021 (Taiwan priority) — *under prosecution*.

INDUSTRY-UNIVERSITY COOPERATIVE RESEARCH PROJECT

1. Robo-advisors in portfolio management, *Synergies Intelligent Systems*. Dec 2016 – Dec 2017

2. Data analytics in finance, *E.Sun Commercial Bank*. Nov 2018 – Jul 2021

3. An Anti-Money-Laundering name-checking solution, *Liu & Partners Law Office*. Jul 2020 – Dec 2020

PRESENTATIONS

Conference and Workshop Presentations

1. "Trading BTC with robust pricing kernels", Digital Finance for Supervision, MSCA DIGITAL Network Workshop, European Central Bank, Frankfurt am Main, Germany (invited; scheduled) Nov 30, 2026
2. "Trading BTC with robust pricing kernels", the 35th South Taiwan Statistics Conference, Department of Statistics and Data Science, National Cheng Kung University, Tainan (invited) Jun 25–26, 2026
3. "Mapping risk contagion through graphical models and early warning signals in Taiwan's stock market" (with Nok Hang Chan and Po-Yao Hsu), invited session EO116 "Graphical and Causal Learning for Complex Stochastic Processes", 9th International Conference on Econometrics and Statistics (EcoSta 2026), Ryukoku University, Kyoto, Japan Aug 8–10, 2026
4. "Trading BTC with robust pricing kernels", Quantitative Finance Conference 2026, Center for Quantitative Finance, National University of Singapore, Singapore (invited) Jun 10–12, 2026
5. "Financial analytics of inverse BTC options in a stochastic volatility world", 2026 AIDA Conference, Bucharest University of Economic Studies, Romania (invited) Jun 6, 2026
6. "Tail Risk in Bitcoin under the Basel Framework" The 19th NYCU International Finance Conference, Hsinchu, Taiwan Dec 5, 2025
7. "Cryptocurrency Risk Under Basel Standards: A Stochastic Volatility Approach with Jumps" 8th International Conference on Econometrics and Statistics (EcoSta 2025) Waseda University, Tokyo, Japan Aug 21–23, 2025
8. "Financial analytics of BTC inverse options in a stochastic volatility world" 2024總體經濟計量模型研討會, Macroeconometric Modelling Workshop, MMW2024, Institute of Economics, Academia Sinica, Taipei, Taiwan Dec 11–12, 2024
9. "Financial Risk Meters in Taiwan's High-Cap Sector", The 32nd Annual PBFEAM, Rutgers Business School - New Brunswick, NJ, USA Aug 28–29, 2024
10. "Financial Risk Meters in Taiwan's High-Cap Sector" 第三十三屆南區統計研討會, Kaohsiung, Taiwan Jun 27–28, 2024
11. "Financial Risk Meters in Taiwan's High-Cap Sector" The 2024 International Conference of TFA, 2024年臺灣財務金融學會年會暨國際研討會, Kaohsiung, Taiwan Jun 7–8, 2024
12. "Financial Analytics of Deribit BTC Options" The 31st Annual Conference on Pacific Basin Finance, Economics, Accounting and Management (PBFEAM) 第三十一屆亞太財務經濟會計與管理研討會, NYCU, Hsinchu Jun 2, 2023
13. "透過群表現理論的一個新穎的球狀蒙地卡羅估計量來計算高維度積分" 第18屆台灣作業研究學會年會暨學術研討會(ORSTW 2022), NYCU, Hsinchu Dec 9, 2022
14. Discussant of "Volatility, Liquidity, ESG Investment and Conditional Value at Risk: Evidence from Taiwanese Financial Holding Companies" 台灣經濟學會2022年年會 Dec 3, 2022
15. "比特幣倒數型選擇權在隨機波動度下的動態避險" 2022 New Futures 期貨學術與實務交流研討會, 臺灣期貨交易所與工商時報共同舉辦, Taipei Nov 28, 2022
16. "Crypto currency returns", the Crypto Currency Day of the CAS Blockchain, on-line presentation, University of Zurich Oct 1, 2022

17. "Pricing and hedging inverse BTC options", 2022 International Conference of the Taiwan Finance Association (TFA 2022), 「期交所期貨與選擇權成果發表會」,國立陽明交通大學資訊管理與財務金融學系 Jun 17, 2022

18. "Pricing and hedging inverse Bitcoin options", the 5th International Conference on Econometrics and Statistics (EcoSta 2022), online presentation, Ryukoku University, Kyoto, Japan (invited)Jun 4–6, 2022

19. "Pricing and hedging inverse Bitcoin options", online presentation, the 30th Annual Conference on Pacific Basin Finance, Economics, Accounting (PBFEAM), National Chung Cheng University, Taiwan (invited) May 13–15, 2022

20. "Pricing and hedging the inverse option under the SVCJ model", online presentation, the 3rd Berlin Conference Crypto-Currencies in a Digital Economy, Berlin, Germany (invited) Nov 26–27, 2021

21. "金融/法律科技", 2021科技與法律跨領域寒假營隊, 財團法人理律文教基金會, Taiwan Feb 2, 2021

22. "On a projection estimator for Monte Carlo integration", 12th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2019), University of London, UK Dec 14–16, 2019

23. "Machine learning for predicting default of credit card holders and success of kickstarters", 11th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2018), University of Pisa, Italy Dec 14–16, 2018

24. "A spherical Monte Carlo approach for calculating Value-at-Risk in financial risk management", the 2017 Winter Simulation Conference, Las Vegas, Nevada, USA Dec 3–6, 2017

25. "A novel spherical Monte Carlo method via group representation", the Seventh Institute for Mathematical Statistics - Finance, Probability and Statistics Workshop, University of Maryland Baltimore County, Baltimore, MD, USA (invited) Jul 27, 2017

26. "Efficient simulation of high-dimensional integrals with applications in finance", NCTS Workshop on Frontier Technologies for High Dimensional Problems and Uncertainty Quantification, National Taiwan University, Taipei, Taiwan Dec 16, 2016

27. "A spherical Monte Carlo method for financial option pricing", the Fifth Institute for Mathematical Statistics - Finance, Probability and Statistics Workshop, Rutgers University, New Brunswick, NJ, USA (invited) Jun 26, 2015

28. "Efficient importance sampling for false alarm probability under K-distributed sea clutter", the Fifth International Workshop in Sequential Methodologies 2015, Columbia University, New York, NY, USA (invited) Jun 23, 2015

29. "Optimal exponentially tilted importance sampling in pricing financial derivatives", 2014財務數學與財務統計研討會, National Taipei University, Taipei, Taiwan Aug 29, 2014

30. "On spherical Monte Carlo simulations for multivariate normal probabilities", the Fourth Institute of Mathematical Statistics - Finance, Probability, and Statistics Workshop, University of Technology Sydney, Sydney, Australia (invited) Jul 3, 2014

31. "A direct method for calculating Greeks under Lévy processes", The 3rd Institute of Mathematical Statistics Asia Pacific Rim Meeting, Taipei, Taiwan Jul 1, 2014

32. "A direct method for calculating Greeks under Lévy processes", the 8th World Congress of the Bachelier Finance Society, Brussels, Belgium Jun 1, 2014

33. "Optimal exponentially tilted importance sampling for event simulation with applications", The 7th NCTU International Finance Conference and Celebration for Professor Cheng-Few Lee's Forty-year Teaching Career, National Chiao-Tung University, Hsinchu, Taiwan Jan 10, 2014

34. "State Price Densities implied from weather derivatives", The 14th Frontier Science Symposium 第十四屆尖端科學研討會, Nanjing, China (invited) Oct 21, 2013

35. "Bayesian analysis and application", 第二十二屆南區統計研討會暨2013年中華機率統計學會年會及學術研討會, National University of Kaohsiung, Kaohsiung, Taiwan Jun 29, 2013

36. "Efficient option pricing with importance sampling", the 4th Conference on Financial Engineering Theory and Practice 第四屆財務工程理論與實務研討會, Providence University, Taichung, Taiwan May 17, 2013

37. "SPDs implied from weather options", the Pricing Kernel Puzzle Workshop, The Konstanz University, Konstanz, Germany (invited) Jul 12, 2011

38. "Greeks of financial options", 第二十屆南區統計研討會暨2011年中華機率統計學會年會及學術研討會, National Chung Cheng University, Chiayi, Taiwan Jun 24, 2011

39. "Bayesian nonparametric approaches for financial option pricing", the 2nd Conference on Financial Engineering Theory and Practice 第二屆財務工程理論與實務研討會, Providence University, Taichung, Taiwan May 16, 2011

40. "Greeks of financial options", 2010開南大學「金融市場與風險管理」研討會, Kainan University, Taoyuan, Taiwan Dec 23, 2010

41. "Bayesian nonparametric approaches for financial option pricing", 2010 中國統計學社社員大會暨國際統計學術研討會, National Central University, Taoyuan Dec 17, 2010

42. "Implied state price densities", Workshop on Pricing Weather And Temperature, National Central University, Taoyuan, Taiwan Dec 6, 2010

Invited Seminars

1. "Cross-Attentive Multimodal RNNs for Forecasting Bitcoin Realized Volatility", Data Science Degree Program, National Taiwan University, Taipei May 29, 2026

2. "Cross-Attentive Multimodal RNNs for Forecasting Bitcoin Realized Volatility", College of Commerce joint seminar (Finance, International Business, and Money & Banking), National Chengchi University, Taipei Apr 24, 2026

3. "Cross-Attentive Multimodal RNNs for Forecasting Bitcoin Realized Volatility", Department of Quantitative Finance, National Tsing Hua University, Hsinchu Mar 19, 2026

4. "Tail Risk in Bitcoin under the Basel Framework", Institute of Statistics, National University of Kaohsiung, Kaohsiung Oct 22, 2025

5. "Bridging Interpretability And Performance: Augmented Features And PowerSHAP For Credit Default Prediction", Department of Economics, National Cheng Kung University, Tainan Jun 10, 2025

6. Financial analytics of BTC inverse options in a stochastic volatility world, Department of Statistics, National Chengchi University, Taipei Jun 2, 2025

7. "Financial analytics of BTC inverse options in a stochastic volatility world", Department of Quantitative Finance, National Tsing Hua University, Hsinchu May 8, 2025
8. "Integrating Explainable AI with Polynomial Analytics to Enhance Credit Scoring Model Compliance", Institute of Statistical Science, Sinica, Taipei Feb 26, 2024
9. "On pricing and hedging inverse BTC options", 國立高雄大學統計學研究所, Kaohsiung Sep 14, 2022
10. "Statistics, AI, freedom of wealth?", 國立陽明交通大學百川學堂, Hsinchu, Taiwan Mar 9, 2022
11. "Calculating Risk Measures and Pricing Financial Derivatives Using Spherical Monte Carlo Simulation", Department of Quantitative Finance, National Tsing Hua University, Hsinchu May 30, 2019
12. "Calculating Risk Measures and Pricing Financial Derivatives Using Spherical Monte Carlo Simulation", Institute of Statistical Science, Academia Sinica, Taipei Mar 18, 2019
13. "A spherical Monte Carlo approach with applications in finance", Department of Statistics, Chinese Hong Kong University, Hong Kong Jul 13, 2017
14. "A novel spherical Monte Carlo method via group representation", Department of Applied Mathematics, National Sun Yat-Sen University, Kaohsiung Sep 25, 2015
15. "Optimal exponentially tilted importance sampling in pricing financial derivatives", Department of Applied Mathematics, National Chiao-Tung University, Hsinchu Oct 21, 2014
16. "蒙地卡羅方法與統計應用", 第五屆聯電夏令數學資優營, National Kaoshiung Normal University, Kaohsiung Aug 24, 2014
17. "A direct method for calculating Greeks under Lévy processes", Institute of Statistics at National Chiao-Tung University, Hsinchu May 23, 2014
18. "On spherical Monte Carlo simulations for multivariate normal probabilities", Department of Statistics, National Cheng-Kung University, Tainan Nov 14, 2013
19. "Overview on weather derivatives", Department of Atmospheric Sciences, National Taiwan University, Taipei Nov 7, 2013
20. "Ideas of kernel functions and applications to asset pricing", Colloquium of Shih-Hsin University Center for Computation and Empirical Statistical Studies, Taipei Sep 26, 2013
21. "Implied state price densities of weather derivatives", Am Weierstrass-Institut fuer Angewandte Analysis und Stochastik sprechen im Rahmen des Forschungsseminars Mathematische Statistik, Humboldt University, Berlin, Germany Jul 11, 2012
22. "Implied state price densities", Department of Finance, National Central University, Taoyuan Mar 2, 2012
23. "Bayesian random trees for calibrating State Price Densities of financial options", Department of Statistics, National University of Kaohsiung, Kaohsiung Sep 21, 2011
24. "Experience sharing (mathematics)", 聯電數學資優營, National Central University, Taoyuan Aug 26, 2011

25. "Implied state price densities: Nonparametric Bayesian approaches", Center for Applied Statistics and Economic, Humboldt University, Berlin, Germany Jun 30, 2011
26. "Bayesian nonparametric approaches for financial option pricing", Institute of Statistics, National Chiao-Tung University, Hsinchu May 20, 2011
27. "金融危機後，統計學家能做什麼?" Department of Finance, Kainan University, Taoyuan Apr 20, 2011
28. "Bayesian nonparametric approaches for financial option pricing" Department of Statistics, Tamkang University, New Taipei City Apr 14, 2011
29. "Implied state price densities: Nonparametric Bayesian approaches", Department of Mathematics, the Pennsylvania State University, University Park, PA, USA Feb 1, 2011
30. "Bayesian nonparametric approaches for financial option pricing", College of Statistical Management, Shanghai University of Finance and Economics, Shanghai, China Oct 14, 2010
31. "Greeks of financial options", Seminars in Financial Mathematics and Financial Statistics, Institute of Statistical Science, Academia Sinica, Taipei Sep 3, 2010

PROFESSIONAL SERVICE

Editorial Positions

1. Associate Editor for Digital Finance Oct 2022 – Present
2. Associate Editor for Management and Market Oct 2022 – Present

Conference Scientific Committees

The Third Yushan Conference, NYCU, Hsinchu, Taiwan Dec 8, 2023

The Fourth Yushan Conference, NYCU, Hsinchu, Taiwan Dec 5–6, 2024

Ad Hoc Reviewing (Selected)

Financial Innovation, Management Science, Operations Research, Review of Quantitative Finance and Accounting, International Review of Economics and Finance

TEACHING EXPERIENCE

National Yang Ming Chiao Tung University: Statistics (I & II), Financial Econometrics, Machine Learning & FinTech, Mathematical Statistics, Doctoral Seminar

PROFESSIONAL DEVELOPMENT

English Medium Instruction (EMI) Certificate for University Lecturers, Oxford EMI Training Jun 2024

The Global Colloquium on Participant-Centered Learning (GloColl), Harvard Business School Summer 2025, Winter 2026

ACADEMIC ADVISEES

46 Master's theses supervised (26 at NYCU, 20 at NCU), plus five undergraduate research teams.
⋆: Indicates rewards or publications derived from thesis work.

Current Advisees (as of Oct 2026)

Ph.D.: 李彥璋 Yen-Chang Lee (Information Management and Finance, since Jul 2022); 張哲睿 Jerry (since Feb 2026)

M.S.: 鄭凱鴻 Allen Cheng and 鄭詠融 Martin Cheng (Applied Mathematics, since Feb 2025, co-advised with Prof. Ming-Hsuan Kang); 吳隆傑 Lung-Chieh Wu (Applied Mathematics, since Jul 2025); 彭甫詔, 陳之怡, 連鈞宥, 陳俊錡, 廖承灝 (Information Management and Finance, since Jan 2026)

Executive M.S.: 江文煌 Ares Chiang and 范艾雯 Eva Fan (since Sep 2024); 黃德榮 Deron Huang (since Feb 2025); 陳峻毅 Ian (since Jan 2026)

Undergraduate: 黃以樂, 吳冠霖 (Applied Mathematics, since Feb 2025); 藍允澤, 馮品維, 歐佳俞 (Information Management and Finance, since Aug 2025); 施又睿 (since Jan 2026); 宮珮媛, 湯以箴, 徐嘉謙, 王冠築 (since Feb 2026); 林羿辰 (since Mar 2026)

National Yang Ming Chiao Tung University

1. 邱翰奕, 用於預測比特幣已實現波動率之跨注意力多模態循環神經網路 Jul 2026
   Han-Yi (Elvis) Chiou, Cross-Attentive Multimodal RNNs for Forecasting Bitcoin Realized Volatility
   Institute of Business and Management (MBA), co-advised with Prof. Hui-Min Chung
   ⋆ **Huei-Wen Teng** and Chiou [2026], *Finance Research Letters*, accepted; 2026 第19屆崇越論文大賞 佳作論文獎

2. 王惠芯, 人機決策之可解釋與成本敏感信用卡盜刷偵測框架 Aug 2026
   Chrissy Wang, An Explainable and Cost-Sensitive Human-in-the-Loop Framework for Credit Card Fraud Detection
   Degree Program of Global Business Administration

3. 蘇家灝 Jun 2026
   Ka-Ho (Ben) Soo, A Hybrid Deep Learning–Association Rule Model for Forecasting S&P 500 Sectors Returns
   Department of Information Management and Finance

4. 余柔柔, 新聞嵌入、UMAP壓縮與企業信用評等預測：一種多模態方法 Jul 2026
   Rou-Rou (Mabel) Yu, News Embeddings, UMAP Compression, and Corporate Credit Rating Prediction: A Multimodal Approach
   Department of Information Management and Finance

5. 陳諾恒, 透過圖模型繪製台灣股市風險傳染及其預警信號 Jul 2025
   Nok Hang Jason Chan, Mapping Risk Contagion through Graphical Models and Early Warning Signals in Taiwan's Stock Market
   Department of Information Management and Finance

6. 黃馨霈, 基於巴塞爾協議之比特幣風險分析：隨機波動與跳躍擴散模型的應用 Jun 2025
   Hsin-Pei Wendy Huang, Evaluating Tail Risk in Bitcoin Using Stochastic Volatility and Jump Models under Basel Standards
   ⋆ **Huei-Wen Teng**, Huang, and Shih [2025], *Finance Research Letters*

7. 施昱全, 質化洞察與量化預測：GPT-4o Mini 預測台灣企業盈餘 Jun 2025
   Yu-Chuan James Shih, Qualitative Insights Meet Quantitative Forecasts with GPT-4o Mini for Taiwan Earnings Prediction
   ⋆ **Huei-Wen Teng**, Huang, and Shih [2025], *Finance Research Letters*; **Huei-Wen Teng**, Shih, and Kang [2026b], *Journal of the Chinese Statistical Association* (forthcoming)

8. 陳怡仁, 融合可解釋性與效能之信用違約預測：增強特徵與 PowerSHAP 方法 Jul 2025
   Yi-Jen Paul Chen, Bridging Interpretability and Performance: Augmented Features and PowerSHAP for Credit Default Prediction

9. 吳柏賢, 重新設計太陽能契約以釋放長期 ESG 投資潛力 Jun 2025
   Po-Hsien Howard Wu, Redesigning Solar Contracts to Unlock Long-Term ESG Investment

10. 陳彥廷 Jul 2024
    Yen-Ting Chen, Exploring Multifaceted Drivers for Bitcoin Return Analysis
    Department of Information Management and Finance

11. 鄭翔澧, 臺灣高市值產業的金融風險指標 Jul 2024
    Siang-Li Jheng, Financial Risk Meters in Taiwan's High-Cap Sectors
    Department of Information Management and Finance

12. 吳旻修, 應用強化學習於台灣股票市場與比特幣交易 Jul 2024
    Min-Hsiu Wu, Applying Reinforcement Learning for Trading in the Taiwan Stock Market and Bitcoin
    Department of Information Management and Finance

13. 林威承, 以粒子群優化法求解用於加密貨幣指數追蹤之基數限制投資組合最佳化問題 Dec 2024
    Wei-Cheng Lin
    Department of Industrial Engineering and Management, co-advised with Prof. Chun-Cheng Lin

14. 潘俞綺, 估計比特幣選擇權之定價核：新冠疫情前後分析 Jul 2023
    Yu-Ci Pan, On BTC Option Pricing Kernels: Pre- and Post-Pandemic Analysis
    Department of Information Management and Finance

15. 張詠淇, 利用 Heston隨機波動率模型進行比特幣倒數型選擇權的定價與避險 Aug 2022
    Yung-Chi Tom Chang, Pricing and hedging inverse BTC options with Heston's stochastic volatility model
    Department of Applied Mathematics, co-advised with Prof. Ming-Hsuan Kang
    ⋆ 榮獲2022 New Futures 期貨學術與實務交流研討會論文金質獎, Chang, **Huei-Wen Teng**, and Härdle [2023], *Journal of Futures and Options*

16. 陳律翔, 全球資產管理利用Black-Litterman模型於臺灣美國股票市場 Aug 2022
    Lu-Xiang Jay Chen, Global portfolio management with the Black-Litterman model using stocks in Taiwan and the US markets
    Department of Applied Mathematics, co-advised with Prof. Ming-Hsuan Kang

17. 李亦涵, 分群後分類方法於二元分類不平衡問題:應用於信用評分模型 Aug 2022
    I-Han Ian Lee, Cluster-then-Predict for Binary Classification with Imbalance Problem: Applications of Credit Scoring Models
    Institute of Data Science and Engineering, co-advised with Prof. Chu-Lan Kao
    ⋆ **Huei-Wen Teng**, Kang, Lee, and Bai [2024], *International Review of Financial Analysis*

18. 呂映蓁, 台灣期交所黃金選擇權的實證分析 Aug 2022
    Ying-Rou Lu, Empirical analysis for gold options in Taiwan Futures Exchange
    Department of Information Management and Finance

19. 張尚文, L-各向同性多維常態分佈之最大概似估計與其應用 Aug 2021
Shang-Wen Jonathan Chang, Maximum Likelihood Estimate of L-Isotropic Multivariate Normal Distributions and Its Applications
Department of Applied Mathematics, co-advised with Prof. Ming-Hsuan Kang
⋆ **Huei Wen Teng**, Li, and Chang [2020], *2020 International Conference on Pervasive Artificial Intelligence (ICPAI)*

20. 李育賢, 傳統方法與機器學習在資產定價上的實證比較 Jun 2020
Yu-Hsien Bill Li, An Empirical Comparison between Traditional Methods And Machine Learning in Asset Pricing Models
Department of Information Management and Finance
⋆ **Huei Wen Teng**, Li, and Chang [2020], *2020 International Conference on Pervasive Artificial Intelligence (ICPAI)*; **Huei-Wen Teng** and Li [2023], *Digital Finance*

21. 吳哲杭, 多元GARCH模型與投資組合管理在上海證交所A股的實證比較 Jun 2020
Zhe-Hang Jeff Wu, An empirical comparison of multivariate GARCH models with a portfolio management application for the A-share in Shanghai Stock Exchange
Department of Information Management and Finance

22. 趙祥安, Heston隨機波動率模型與NGARCH選擇權定價模型在匯率連動選擇權上之實證比較 Jun 2020
Xiang-An Zhao, An empirical comparison between the Heston stochastic volatility model and the NGARCH model in pricing quanto options
Department of Applied Mathematics, co-advised with Prof. Ming-Hsuan Kang
⋆ **Huei-Wen Teng** and Zhao [2021], *Journal of Futures and Options*

23. 林益嘉, 基於深度學習之時間序列因子於信用評級模型 Jun 2019
Yi-Chia Kelly Lin, Deep time-series feature extraction in credit scoring models
Department of Information Management and Finance

24. 許至妤, 基於巴塞爾協議在金融風險管理中 VaR 估計的實證研究 Jun 2019
Chih-Yu Emma Hsu, An empirical investigation on estimations of VaR in financial risk management based on Basel Accords
Department of Information Management and Finance

25. 陳奕翔, 機器學習中的Boosting方法及其在信用風險管理違約預測中的應用 Jul 2019
Yi-Xiang Arikson Chen, On Boosting Methods in Machine Learning with Applications in Delinquency Prediction for Credit Risk Management
Department of Information Management and Finance

26. 黃春僖, 分群技術在大型資產配置的應用 Jun 2019
Chun-Xi Simon Huang, Large portfolio management with clustering techniques
Department of Applied Mathematics, co-advised with Prof. Ming-Hsuan Kang

Graduate Institute of Statistics, National Central University

1. 葉惠瑄 Jun 2017
Huei-Hsuan Yeh, A Multivariate Markov Switching Model for Portfolio Optimization

2. 應劭玄, 在馬可夫轉換模型下的資產配置 Jun 2017
Shao-Hsuan Jason Yin, Portfolio Allocation with Regime Switching Models

3. 李宛柔 Jun 2017
Wan-Zoe Cora Lee, A Dynamic Rebalancing Strategy for Portfolio Allocation

4. 吳柏辰 Jul 2016
Po-Chen Wu, Simulating average run lengths of a copula-based control chart with the use of control variates

5. 江厚德 Jul 2016
   Arron Chiang, An empirical comparison of various approaches in calculating Value at Risk

6. 林煒紘 Jun 2015
   Wei-Hung Lin, Asset allocation based on the Black-Litterman and GARCH models
   ⋆ Lin, **Huei-Wen Teng**, and Yang [2020], *Handbook of Financial Econometrics, Mathematics, Statistics, and Machine Learning*

7. 沈睿謙, 在Black-Scholes模型下運用選擇權資料進行動態避險之比較 Jun 2015
   Rui-Qian Shen, Comparisons of dynamic hedging of financial options using different volatility estimators under the Black-Scholes model

8. 楊舒媛 Jun 2015
   Shu Yuang Yang, Modelling the VIX index and hedging the S&P 500 futures using VIX options

9. 王聖翔 Jun 2014
   Sheng-Xiang Wang, A direct method for calculating Greeks under some L'evy processes
   ⋆ Lyuu, **Huei-Wen Teng**, Tseng, and Wang [2019], *Quantitative Finance*

10. 曾耀德 Jun 2014
    Yao-Te Tseng, Sensitivity analysis of credit derivatives
    ⋆ Lyuu, **Huei-Wen Teng**, Tseng, and Wang [2019], *Quantitative Finance*

11. 李宥萱 Jun 2014
    Yu-Hsuan Lee, Improved mortality forecasting using Augmented Data
    ⋆ Luo, **Huei-Wen Teng**, and Lee [2016], *Asia-Pacific Journal of Risk and Insurance*

12. 陳彥勳 Jun 2013
    Yen-Hsun Chen, Structure learning for hierarchical Archimedean copulas

13. 沈志泰 Jun 2013
    Chih-tai Shen, Estimating intensity processes from credit default swaps

14. 吳諭昕 Jun 2013
    Yu-Hsin Wu, Calibrating the state price densities using TAIEX options

15. 陳醇潔 Jun 2012
    Chun-Chieh Chen, Efficient option pricing with importance sampling
    ⋆ 榮獲100學年度中國統計學社論文獎優等; Chen, Fuh, and **Huei-Wen Teng** [2013], *Journal of the Chinese Statistical Association*; **Huei-Wen Teng**, Fuh, and Chen [2016], *Quantitative Finance*

16. 劉家齊 Jun 2012
    Chia-Chi Liu, On pricing credit default swaps

17. 趙彥茹 Jun 2012
    Yen-ju Chao, Bayesian imputation with an application to mass-period functions of extrasolar planets
    ⋆ **Huei-Wen Teng**, Hung, and Chao [2015a], *Journal of Applied Statistics*

18. 陳彥鈞 Jun 2012
    Yen-Chun Chen, Copula-based weather data forecasting

19. 劉佑聖 Jun 2011
    You-Sheng Liu, Portfolio selection based on copula models with applications in Taiwan stock market

20. 吳嘉洋 Jun 2011
    Jia-Yang Wu, Copula-based time series with applications to unemployment rates modeling

Undergraduate Projects

1. 許柏堯 Frank Hsu, 黃以樂 Yile Huang, 吳冠霖 Guan-Lin Wu Spring, Fall 2025
   已嚴肅學習圖模型
   ⋆ 國立陽明交通大學第三屆智慧創新跨域潛力人才競賽,金融科技組,優選

2. 鄭有朋 You-Peng Zheng, 王鳴謙 Ken Wang, 李語涵 Yu-Han Li, 吳珮慈 Patty Wu, 許晉 Chris Spring, Fall 2023
   探索比特幣回報分析的多面向驅動因素
   ⋆ 國立陽明交通大學第一屆智慧創新跨域潛力人才競賽,金融科技組,優等

3. 陳之怡, 唐君亞, 蔡昕妤 Spring, Fall 2023
   台灣高市值電子與金融行業的金融風險計量指標
   ⋆ 國立陽明交通大學第一屆智慧創新跨域潛力人才競賽,金融科技組,優等

4. 陳品蓁 Jasmine Chen, 宋佩芩 Sammi Sung Aug 2021 – Jul 2022
   Statistical modeling for the exchange rate between NTD and USD

5. 范凱翔 Tom Fan (應數系, co-advised with Prof. Ming-Hsuan Kang), 林睿瑜 Phoebe Lin, 陸恭葦 Lucy Lu Aug 2019 – Jul 2020
   Time Series Forecasting in Credit Scoring Model
   ⋆ 林睿瑜榮獲109年度科技部大專生計畫，Time Series Forecasting in Credit Scoring Model，計畫編號：MOST109-2813-C-009-069-H。

GRANTS

NSTC Principal Investigator on 17 projects since 2010 (NTD 15.5 million in total, including a short-term international visitor grant); Co-Principal Investigator on NSTC projects totaling NTD 15.0 million.

Principal Investigator

1. *115* 年度, 銜接實體與風險中性動態：加密貨幣市場定價核分析之雙測度框架研究, 專題研究計畫（一般研究計畫）, 統計方法, 計畫主持人.

2. *115* 年度, 金融市場的尾部風險衡量, 專題研究計畫（雙邊協議型擴充加值(add-on)國際合作研究計畫）, 統計方法, 計畫主持人.

3. *114* 年度, 深究臺灣金融市場之系統性風險：運用金融風險計量器與功能性結構方程模型, 專題研究計畫（一般研究計畫）, 統計方法, 計畫主持人.

4. *114* 年度, 邀請國際科技人士短期訪問：李光堯 Kuang-Yao Lee, 短訪計畫, 統計方法, 計畫主持人.

5. *112* 年度, 金融分析及其在實證資產定價和加密貨幣選擇權定價的應用, 專題研究計畫（一般研究計畫）, 統計方法, 計畫主持人.

6. *111* 年度, 加密貨幣選擇權的定價與避險, 專題研究計畫（一般研究計畫）, 統計方法, 計畫主持人.

7. *110* 年度, 蒙地卡羅模擬方法在傳統財務問題與金融科技, 專題研究計畫（一般研究計畫）, 統計方法, 計畫主持人.

8. *108* 年度, 機器學習與金融科技:大型投資組合管理與違約風險預測, 專題研究計畫（一般研究計畫）, 數理統計與機率, 計畫主持人.

9. *107* 年度, 利用變化點檢測的動態資產配置策略及其他在金融科技上的應用, 專題研究計畫（一般研究計畫）, 統計方法, 計畫主持人.

10. *106* 年度, 模擬高維度積分及其財務上的應用, 專題研究計畫（一般研究計畫）, 統計方法, 計畫主持人.
11. *105* 年度, 一個利用群表現理論的新穎的球狀蒙地卡羅方法, 專題研究計畫（一般研究計畫）, 統計方法, 計畫主持人.
12. *104* 年度, 模擬不偏的導數估計值與財務應用, 專題研究計畫（一般研究計畫）, 數理統計與機率, 計畫主持人.
13. *103* 年度, 球狀蒙地卡羅方法與應用, 專題研究計畫（探索研究計畫）, 數理統計與機率, 計畫主持人.
14. *102* 年度, 金融選擇權的狀態價格密度:多個到期日和多個資產, 專題研究計畫（新進人員研究計畫）, 統計方法, 計畫主持人.
15. *101* 年度, 高維狀態價格密度函數, 專題研究計畫（新進人員研究計畫）, 財務統計(工程、數學), 計畫主持人.
16. *100* 年度, 高維數據建模與財務應用, 專題研究計畫（新進人員研究計畫）, 財務統計(工程、數學), 計畫主持人.
17. *99* 年度, 貝式財務選擇權定價的無母數方法, 專題研究計畫（新進人員研究計畫）, 其他應用統計, 計畫主持人.

Co-Principal Investigator

1. *110* 年度, 人工智慧在資產配置、衍生性商品定價與風險管理的應用 (3/3), 專題研究計畫（一般研究計畫）, 人工智慧(AI)專案計畫（智慧服務領域）, 共同主持人.
2. *109* 年度, 人工智慧在資產配置、衍生性商品定價與風險管理的應用 (2/3), 專題研究計畫（一般研究計畫）, 人工智慧(AI)專案計畫（智慧服務領域）, 共同主持人.
3. *108* 年度, 人工智慧在資產配置、衍生性商品定價與風險管理的應用 (1/3), 專題研究計畫（一般研究計畫）, 人工智慧(AI)專案計畫（智慧服務領域）, 共同主持人.
4. *103* 年度, 風險管理中對於投資組合風險值和信用違約交換的評估, 專題研究計畫（一般研究計畫）, 財務, 共同主持人.

Last updated: October 21, 2026